MAT 3007 – Optimization

Modeling and Linear Optimization

Lecture 03

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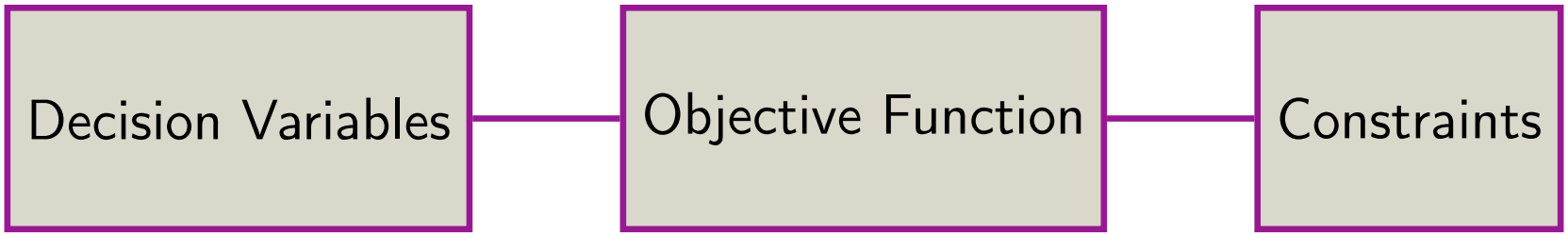
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Recap

Recap: Modeling and Agenda

Identify and formulate the three components:

- Decision $\rightsquigarrow$ Decision variables.
- Objective $\rightsquigarrow$ Objective function.
- Constraints $\rightsquigarrow$ Constraint functions: equalities/inequalities.



Modeling: Support Vector Machines

Support Vector Machines – Setup

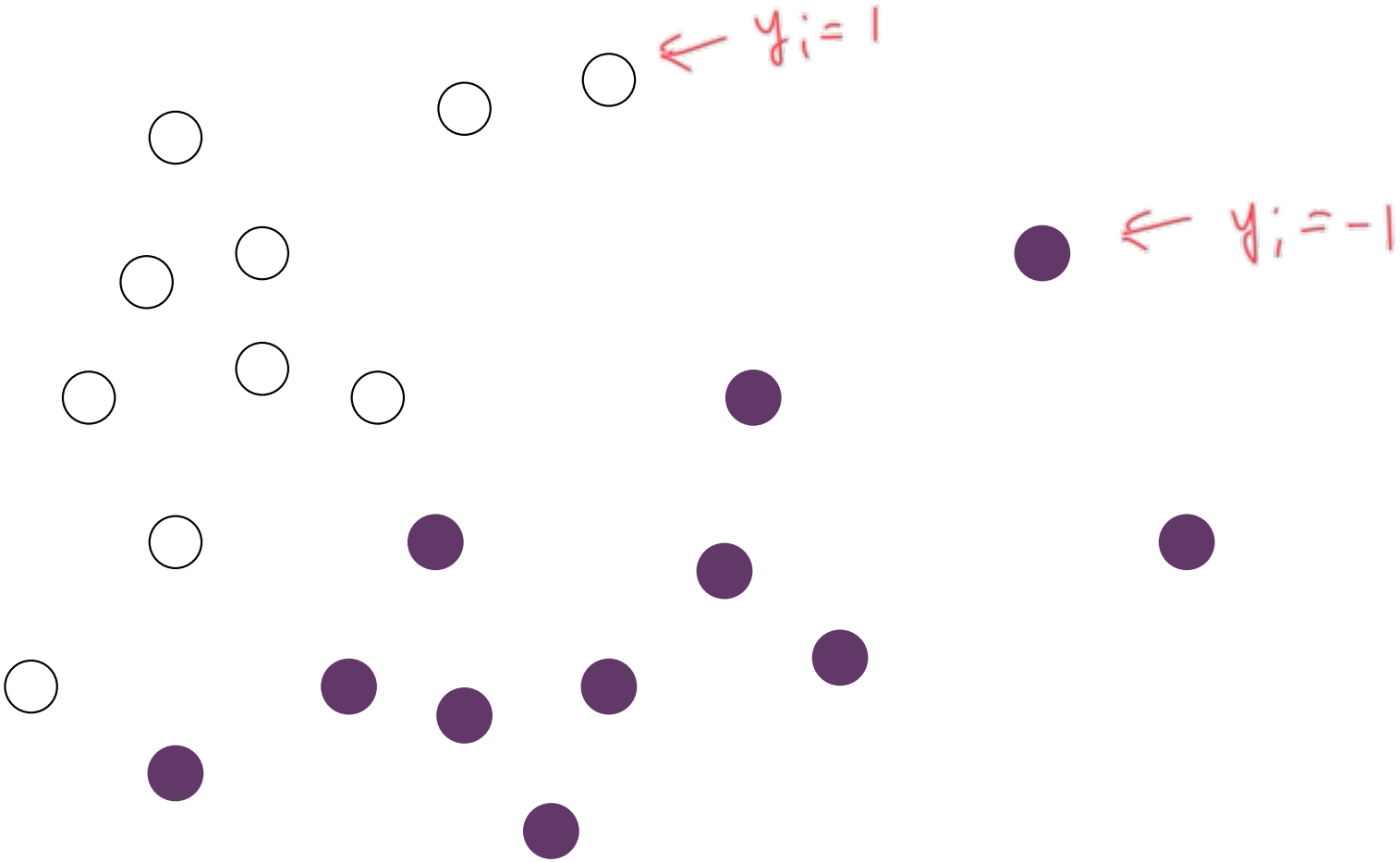
General Setup:

- Given: m objects represented by vectors $x_1, x_2, ..., x_m \in \mathbb{R}^n$ with labels $y_i \in \{-1, 1\}$.
- There are two classes, A and B. The object i belongs to class A if $y_i = 1$ and belongs to B if $y_i = -1$.
- Goal: Learn a function $\ell : \mathbb{R}^n \to \mathbb{R}$ based on the training sample $(x_1, y_1), ..., (x_m, y_m)$, so that when we see a new object z, the function $\ell(z)$ predicts its label/class.

- Example: Blind taste – determine the color from the taste?

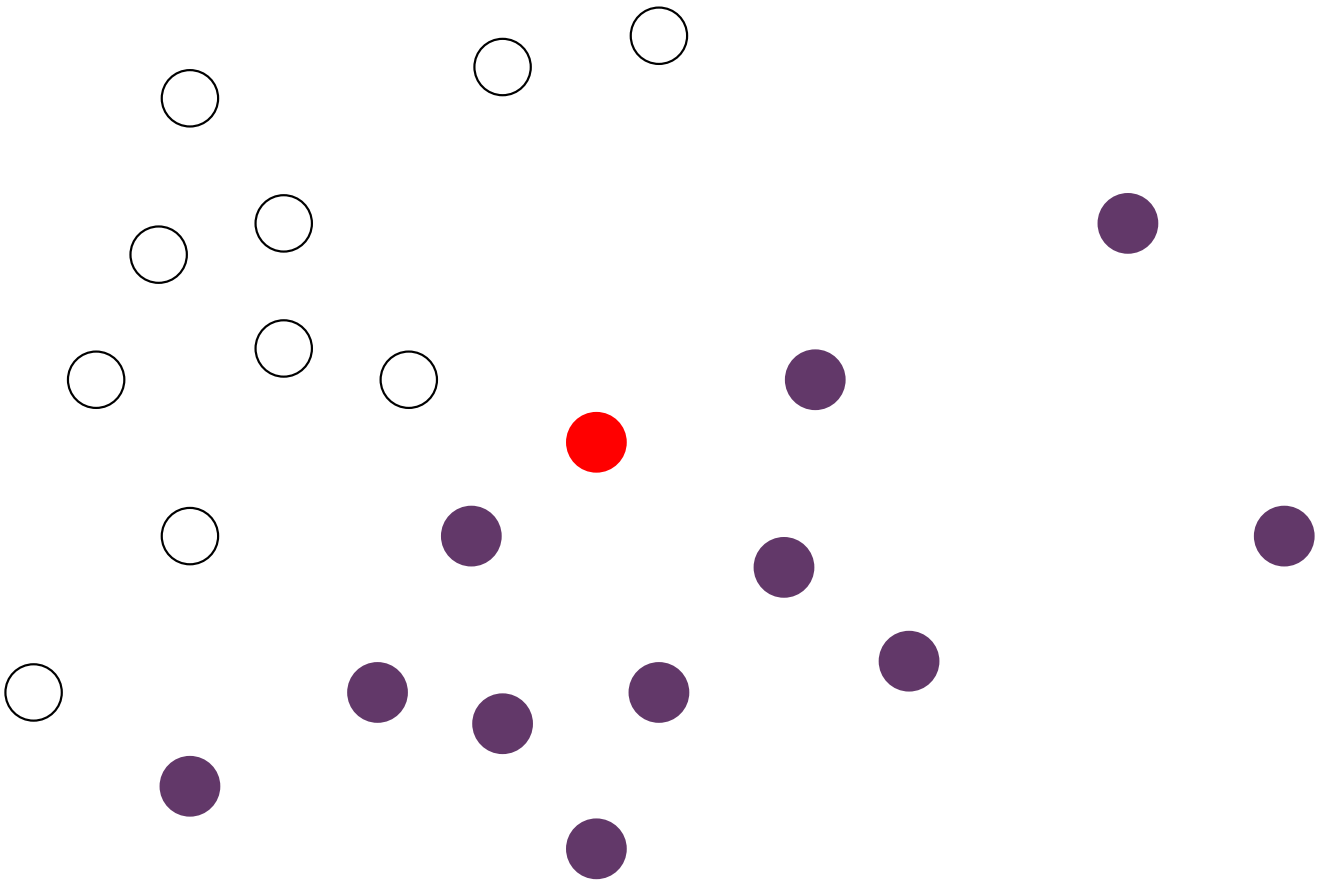
$$x_i \equiv \begin{bmatrix} \text{juicy/fresh} \\ \text{body} \\ \text{acidity} \\ \vdots \end{bmatrix}, \quad y_i = \begin{cases} +1 & \text{white wine,} \\ -1 & \text{red wine.} \end{cases}$$

Support Vector Machines: Illustration



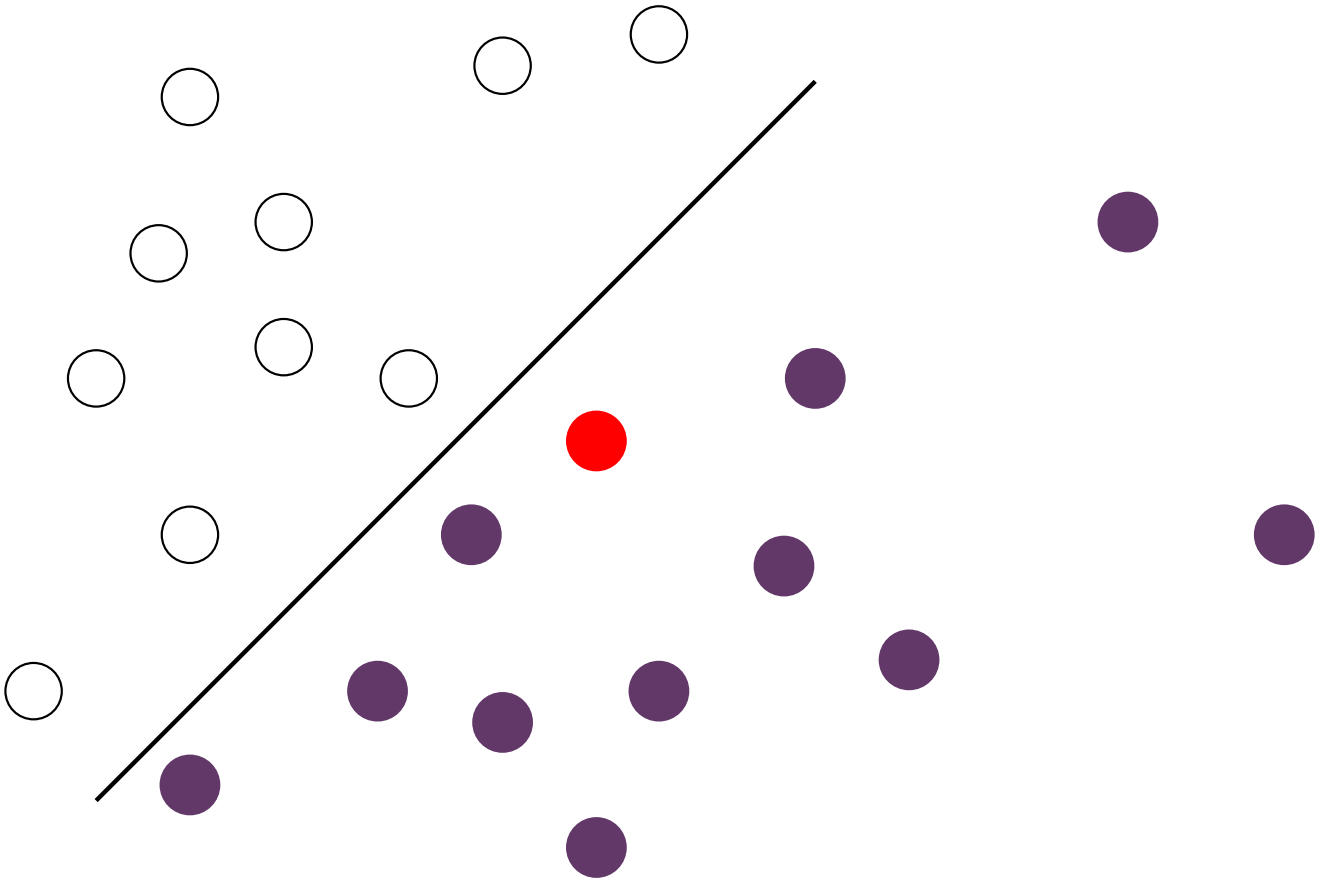
- Consider two labeled sets of points (white and purple).

Support Vector Machines: Illustration



- Consider two labeled sets of points (white and purple).
- $\rightsquigarrow$ Question: Can we predict the label of a newly added point?

Support Vector Machines: Illustration



- Separate the two data sets with a hyperplane!

Assume: there exists a hyperplane $\tilde{\ell}(x) := \tilde{w}^\top x + \tilde{b}$ defined by $(\tilde{w}, \tilde{b})$ that separates the datapoints such that:

$$y_i = \begin{cases} +1 & \text{if } \tilde{\ell}(x_i) > 0, \\ -1 & \text{if } \tilde{\ell}(x_i) \leq 0. \end{cases}$$

Why? we can find $\bar{b}$ s.t. $(\tilde{w}, \bar{b})$ satisfies $y_i = \begin{cases} +1, & \text{if } \tilde{w}^\top x + \bar{b} > a \\ -1, & \text{if } \tilde{w}^\top x + \bar{b} < -a \end{cases}$ for some $a > 0$.

Then there must exist a hyperplane $\ell(x) := w^\top x + b$ defined by (w, b) such that:

$$y_i = \begin{cases} +1 & \text{if } \ell(x_i) \geq +1, \\ -1 & \text{if } \ell(x_i) \leq -1, \end{cases} \quad \forall\ i = 1, \ldots, m.$$

Then we can rescale $(\tilde{w}, \bar{b})$.

Goal: find such (w, b).

To find (w, b) such that $\ell(x) := w^\top x + b$ satisfies

$$y_i = \begin{cases} +1 & \text{if } \ell(x_i) \geq +1, \\ -1 & \text{if } \ell(x_i) \leq -1, \end{cases} \quad \forall\ i = 1, ..., m.$$

observe that this is equivalent to

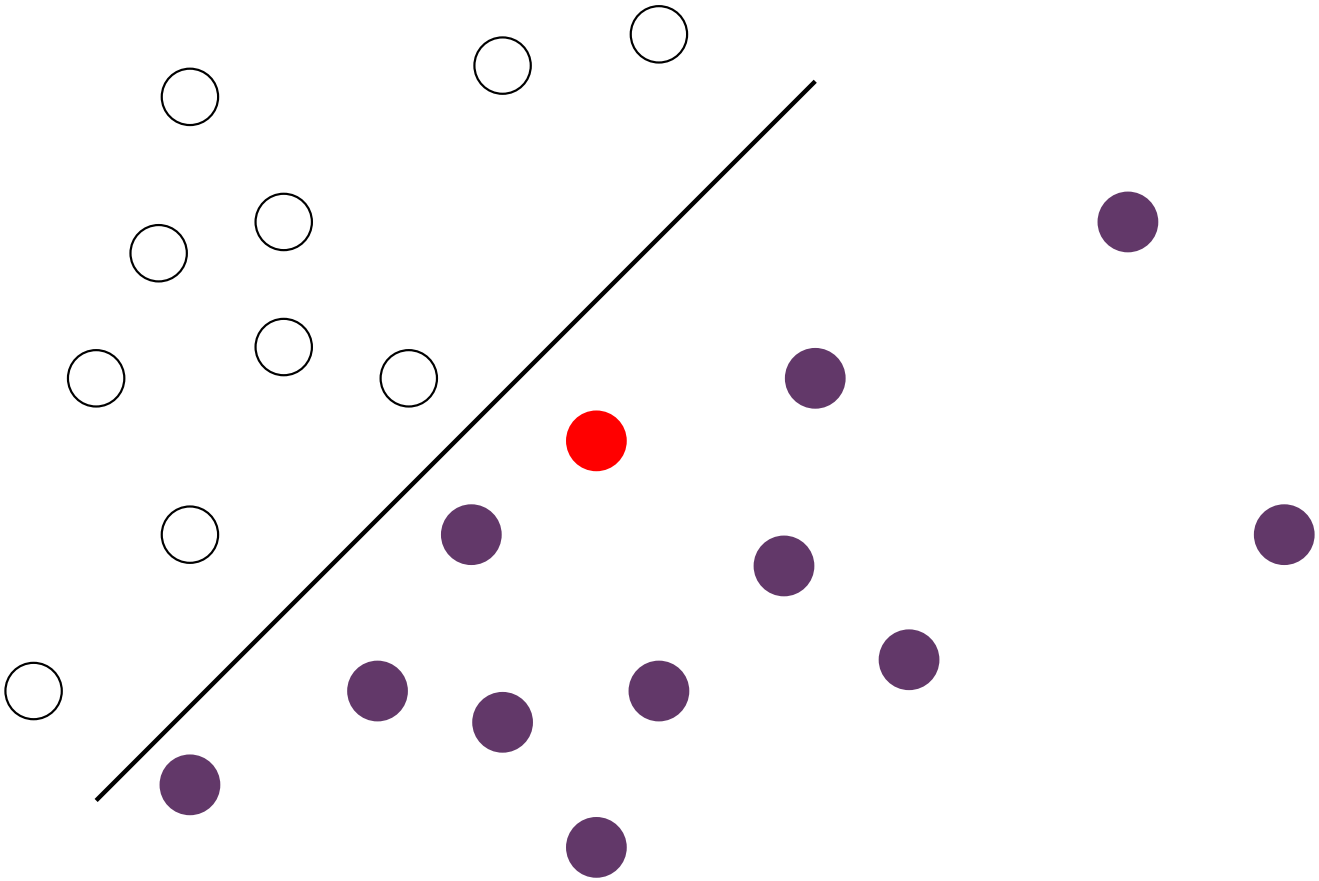
$$y_i\ \ell(x_i) \geq 1.$$

Therefore, we can write the optimization problem as:

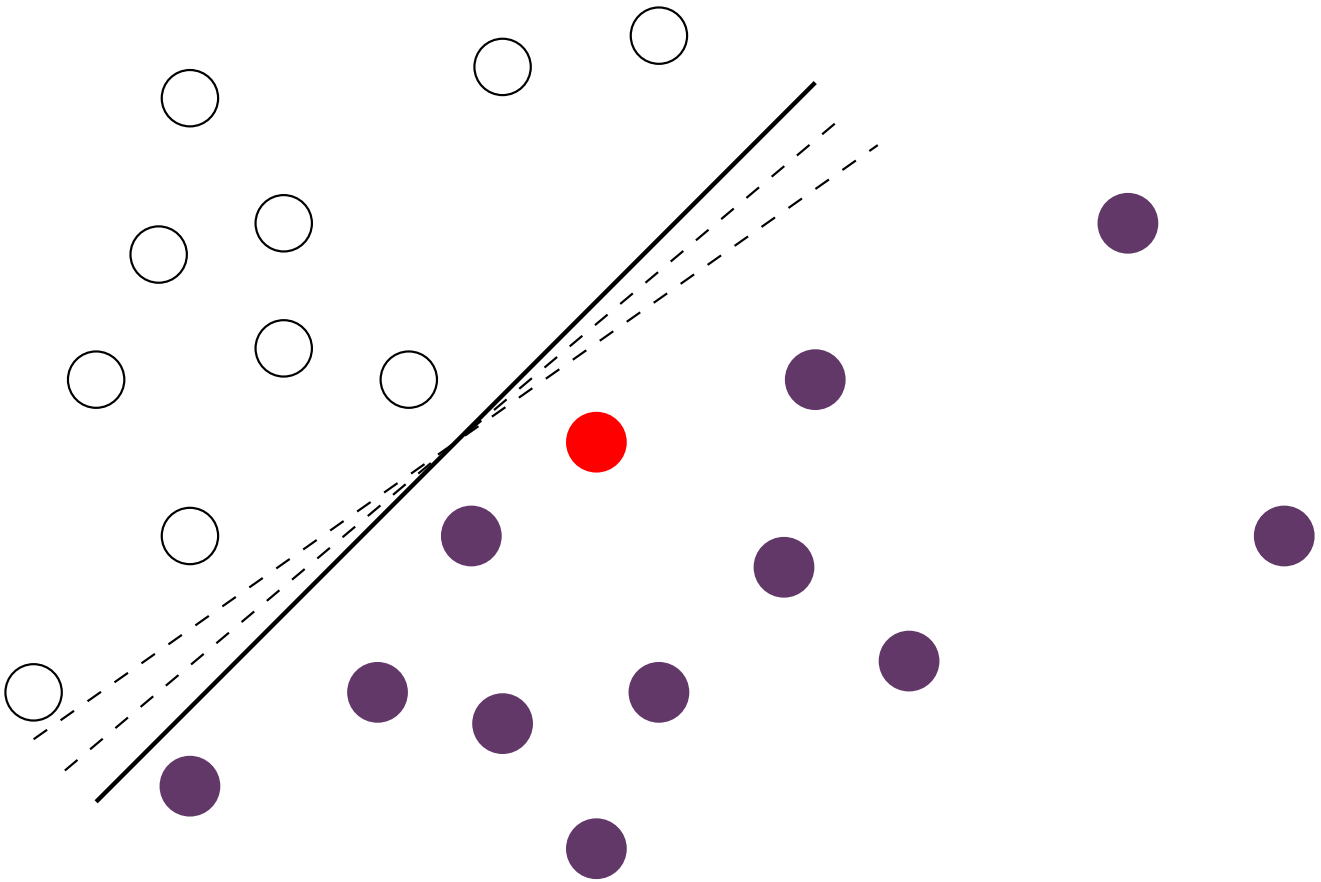
$$\text{minimize}_{w,b}\ 0 \quad \text{s.t.} \quad y_i(x_i^\top w + b) \geq 1, \quad \forall\ i.$$

- This problem is a feasibility problem: feasibility problems are a special kind of optimization problem.

Support Vector Machines: Illustration



Support Vector Machines: Illustration



- Problem: The separating hyperplane might not be unique.

Support Vector Machines: Model Improvement

- Select the "best" hyperplane to separate the two groups.
- Distance between the hyperplanes $\{x : w^\top x + b = 1\}$ and $\{x : w^\top x + b = -1\}$ is $2/\|w\|$ (why?).

Maximize the possible margin (distance between the datasets):

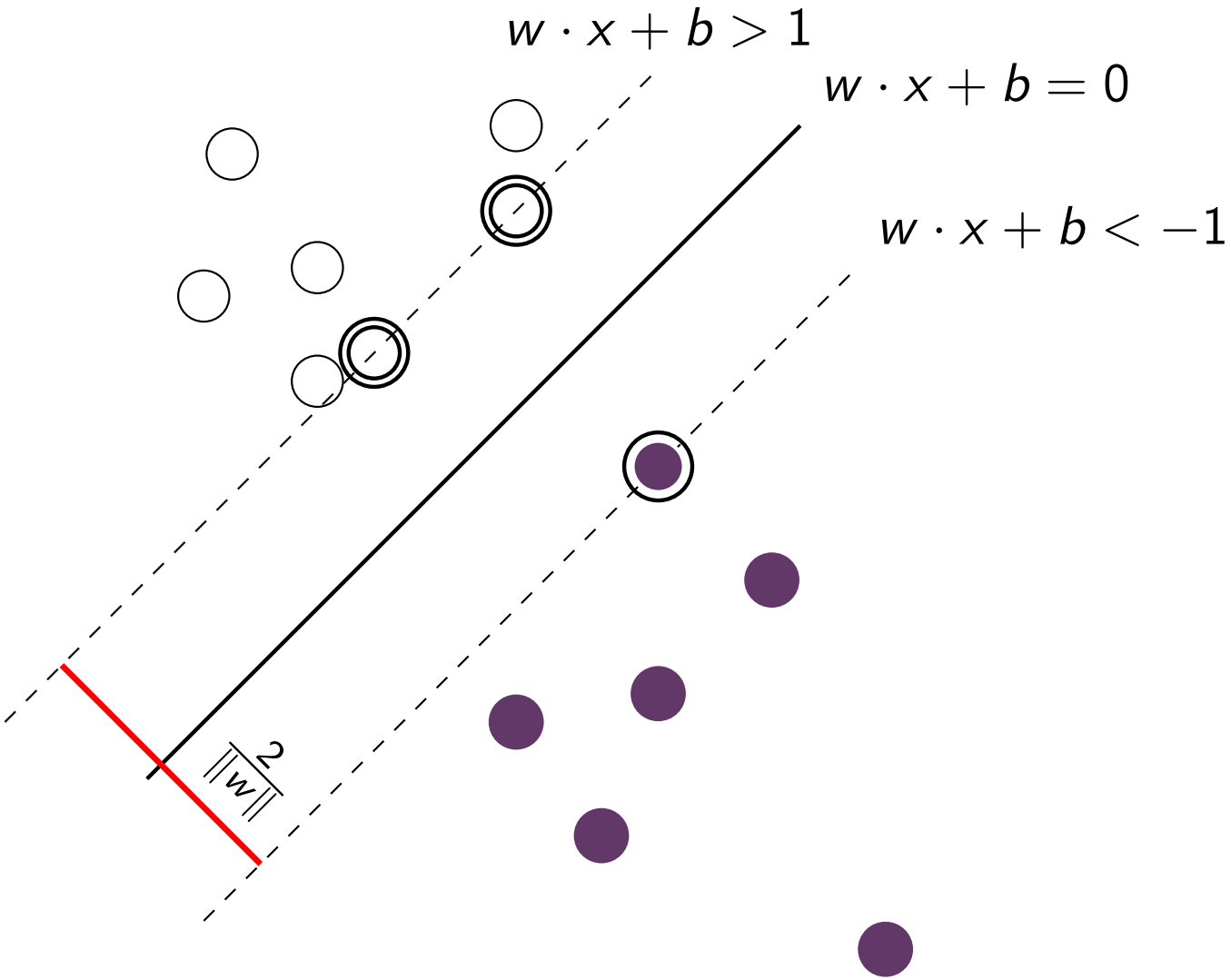
$$\text{maximize}_{w,b} \quad \frac{2}{\|w\|} \quad \text{s.t.} \quad y_i(x_i^\top w + b) \geq 1, \quad \forall\, i.$$

Equivalent formulation (the optimal solutions are the same):

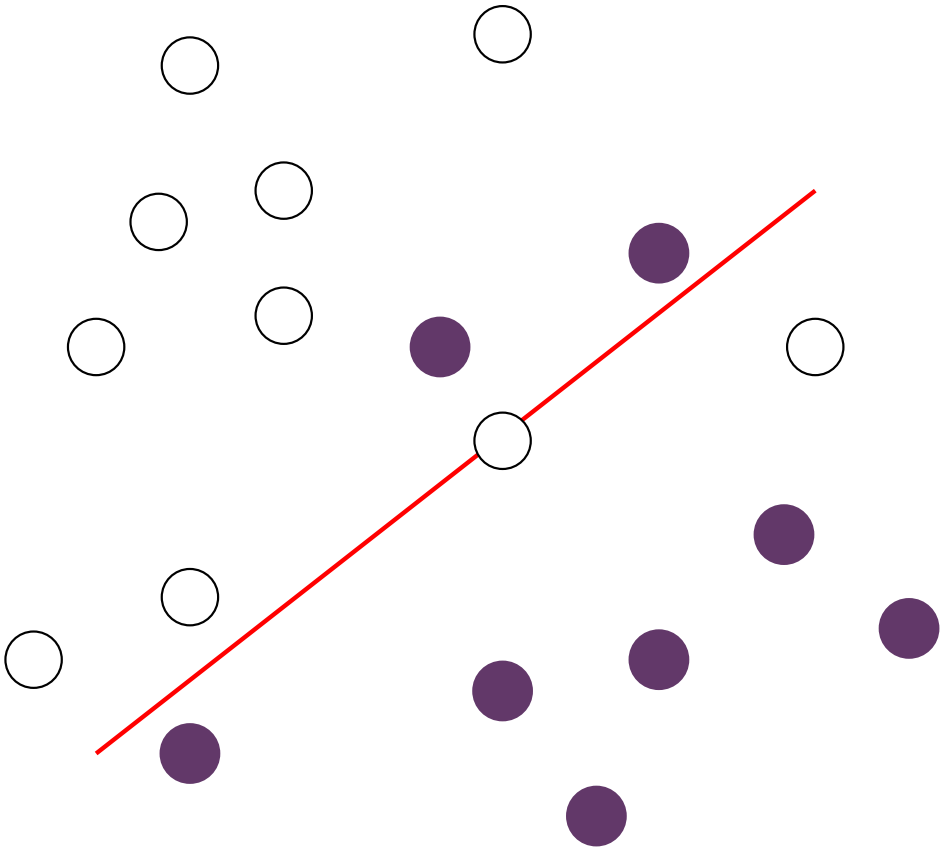
$$\text{minimize}_{w,b} \quad \frac{1}{2}\|w\|^2 \quad \text{s.t.} \quad y_i(x_i^\top w + b) \geq 1, \quad \forall\, i.$$

- We prefer to use $\|w\|^2$ instead of $\|w\|$ because $w \mapsto \|w\|$ is not differentiable at 0.

Support Vector Machines: Illustration



Support Vector Machines: Misclassification



- Question: What to do if the training set can not be perfectly separated by a hyperplane?

Handling Misclassification

Since we may not find (w, b) such that $\ell(x) := w^\top x + b$ satisfies

$$y_i \cdot \ell(x_i) \geq 1, \ \forall i$$

we need to quantify misclassification error. Hinge-Loss is an option

$$\sum_{i=1}^{m} \max\{0, 1 - y_i \ell(x_i)\}.$$

Why is it reasonable?

1. if $y_i \ell(x_i) \geq 1$, then $\max\{0, 1 - y_i \ell(x_i)\} = 0$;
2. if $y_i \ell(x_i) < 1$, then $\max\{0, 1 - y_i \ell(x_i)\} = 1 - y_i \ell(x_i) > 0$.

Strategy and the Full SVM Problem:

The objective combines the margin and misclassification error:

$$\min_{w,b} \quad \frac{\lambda}{2}\|w\|^2 + \sum_{i=1}^{m} \max\{0, 1 - y_i(x_i^\top w + b)\}, \quad \lambda > 0.$$

not linear not linear

- λ is chosen to balance margin and misclassification.
- This is an unconstrained, nonlinear, continuous problem.

A LP Reformulation when $\lambda = 0$

When $\lambda = 0$, the problem becomes

$$\min_{w,b} \quad \sum_{i=1}^{m} \max\{0, 1 - y_i(x_i^\top w + b)\}, \quad \lambda > 0.$$

With notation $x^+ := \max\{0, x\}$, it is equivalent to (in the sense that optimal values are the same, and optimal (w, b) are the same)

$$\begin{aligned} \text{minimize}_{w,b,t} \quad & \sum_i t_i \\ \text{subject to} \quad & t_i = (1 - y_i(x_i^\top w + b))^+, \quad \forall\ i. \end{aligned}$$

(original)

We claim that we can relax "=" to "$\geq$" without changing the optimal solutions and optimal value (why?):

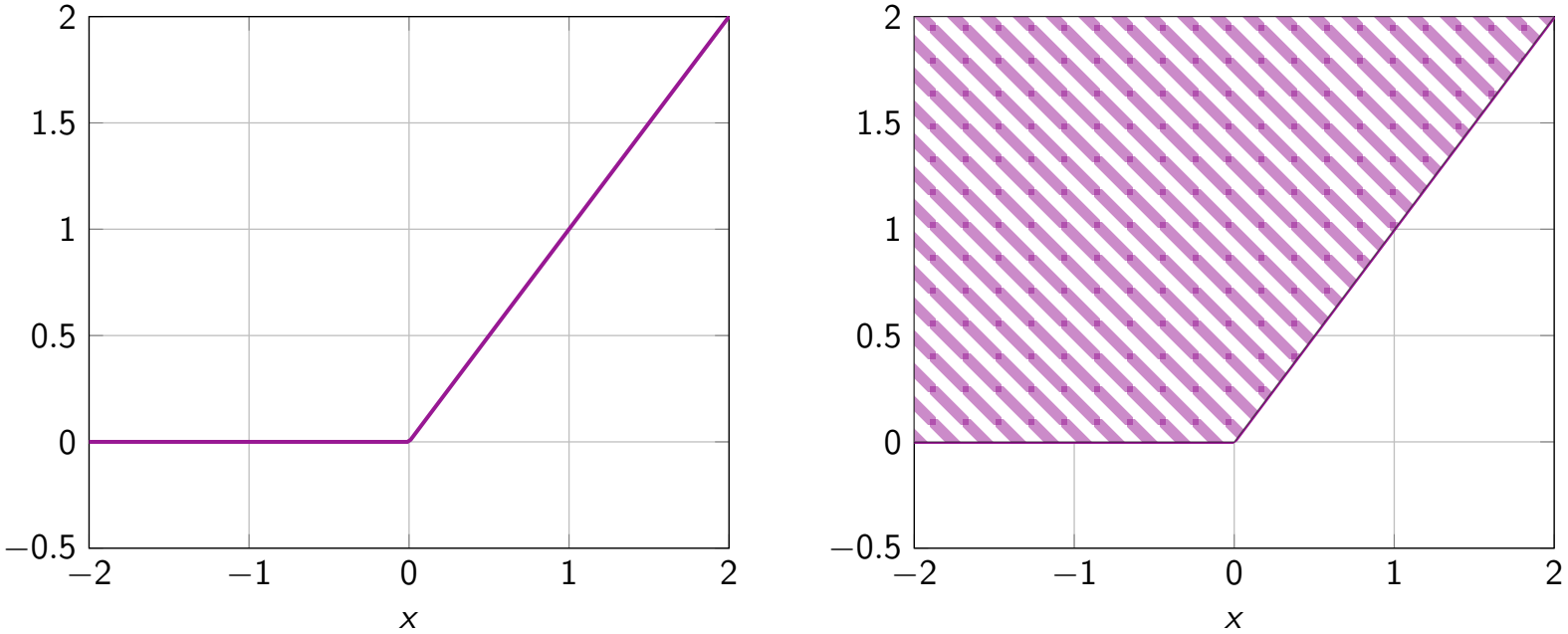
$$\begin{aligned} \text{minimize}_{w,b,t} \quad & \sum_i t_i \\ \text{subject to} \quad & t_i \geq (1 - y_i(x_i^\top w + b))^+, \quad \forall\ i. \end{aligned}$$

(relaxed)

Visualizing the Relaxation "$=$" $\rightarrow$ "$\geq$"

"relax" usaually means that we relax the constraints.

$\max\{0, x\} = t \quad \Longrightarrow \quad \max\{0, x\} \leq t$



- Form minimization problem, the relaxed problem has an optimal value no greater than the original problem.
- Why do we have the same optimal solutions and optimal value after relaxation for this problem?

A LP Reformulation when $\lambda = 0$

Statement: For (relaxed) problem, for any optimal solution (t^*, w^*, b^*) we must have $t_i^* = \left(1 - y_i(x_i^T w^* + b^*)\right)^+$.

Intuition: Since the obj. is $\sum_i t_i$, it will push t_i to choose the smallest possible value.

More rigorous: If $\exists$ an optimal sol. (t^*, w^*, b^*) and $t_i^* > \left(1 - y_i(x_i^T w^* + b^*)\right)^+$ for some i, then we can find another $\tilde{t}_i^*$, e.g., $\tilde{t}_i^* = \frac{1}{2} t_i^* + \frac{1}{2}\left(1 - y_i(x_i^T w^* + b^*)\right)^+$, satifies that $\left(1 - y_i(x_i^T w^* + b^*)\right)^+ \le \tilde{t}_i^* < t_i^*$. Therefore replacing t_i^* with $\tilde{t}_i^*$ leads to smaller obj. $\Rightarrow$ contradiction.

A LP Reformulation when $\lambda = 0$

$x \geq \max\{y, z\} \Leftrightarrow x \geq y$ and $x \geq z$

Furthermore, $t_i \geq (1 - y_i(x_i^\top w + b))^+$ is equivalent to:

$$t_i \geq 1 - y_i(x_i^\top w + b), \qquad t_i \geq 0.$$

Therefore, the optimization problem can be reformulated as:

$$\begin{array}{lll} \text{minimize}_{w,b,t} & \sum_i t_i & \\ \text{subject to} & y_i(x_i^\top w + b) + t_i \geq 1, & \forall\ i \\ & t_i \geq 0 & \forall\ i. \end{array}$$

- This is a linear optimization problem with decision variables $w \in \mathbb{R}^n$, $b \in \mathbb{R}$, and $t \in \mathbb{R}^n$.

Reformulation Steps

What we have done here:

1. Introducing a new auxiliary variable, t.
2. Relaxation of the constraints.

$\rightsquigarrow$ Minimization of the objective function "pushes" the solution towards the desired direction and original constraints.

3. Identifying the correct equivalence.

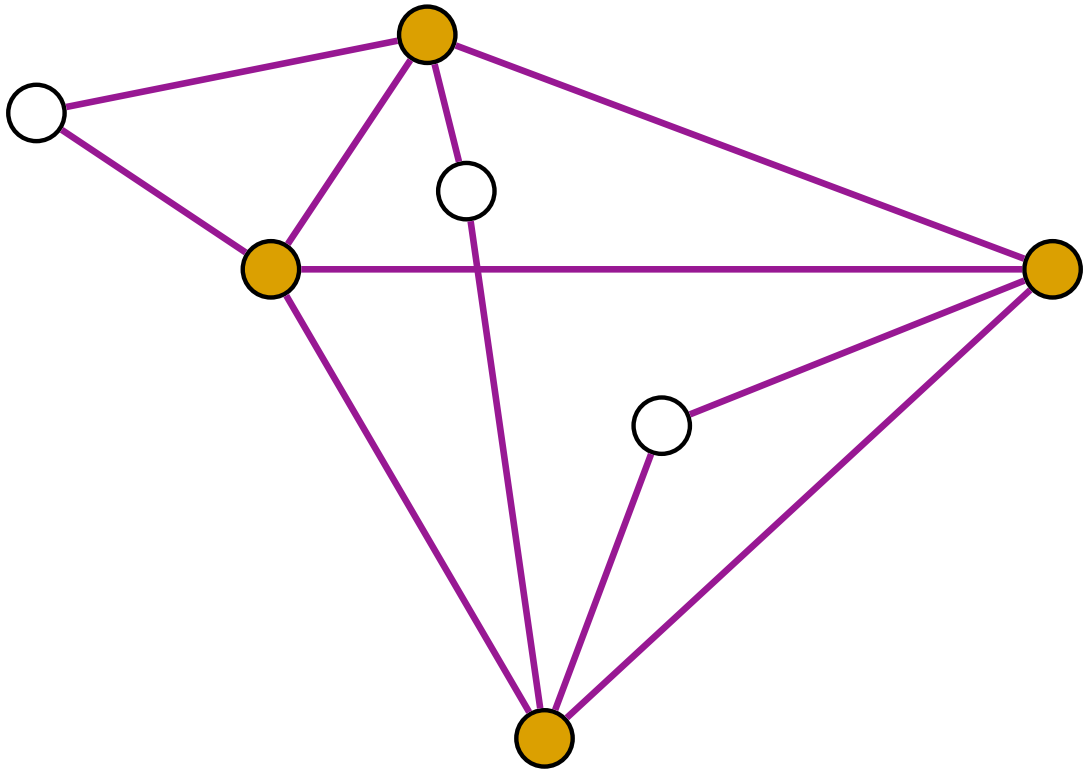
Similar techniques can be applied to many other linear programming reformulation examples.

We should find a pattern for such reformulation as linear programs.

Modeling: Vertex Cover Problem

Another Graph Problem: Vertex Covering

Task: Given a graph with nodes V and edges E, find the smallest set of vertices that touch every edge of the graph.



Optimization Model for Vertex Cover Problem

Define the following decision variables:

$$x_i = \begin{cases} 1 & \text{if we choose vertex } i, \\ 0 & \text{otherwise.} \end{cases}$$

An optimization model can be written as:

$$\begin{aligned} \text{minimize} \quad & \sum_i x_i \\ \text{subject to} \quad & x_i + x_j \geq 1 \quad \forall\, (i,j) \in E \\ & x_i \in \{0,1\} \quad \forall\, i \in V. \end{aligned}$$

- This is an integer (linear) optimization problem.
- Many graph/network problems can be modeled as optimization problems.

Linear Optimization

Definition: Linear Program

Linear Problems

A linear optimization problem or linear program (LP) is an optimization problem in which the objective function and all constraint functions are linear/affine-linear in decision variables.

Example (Production Planning Problem):

$$\begin{aligned} \text{maximize} \quad & x_1 + 2x_2 \\ \text{subject to} \quad & x_1 \leq 100 \\ & 2x_2 \leq 200 \\ & x_1 + x_2 \leq 150 \\ & x_1, x_2 \geq 0. \end{aligned}$$

- Linear optimization is a subclass of continuous optimization.

General Formulation

A linear optimization problem can be generally written as:

$$
\begin{aligned}
\text{minimize/maximize}_{x\in\mathbb{R}^n} \quad & c^\top x \\
\text{subject to} \quad & a_i^\top x \geq b_i \quad \forall\ i = 1, 2, \ldots, m_1 \\
& h_i^\top x \leq d_i \quad \forall\ i = 1, 2, \ldots, m_2 \\
& g_i^\top x = e_i \quad \forall\ i = 1, 2, \ldots, m_3 \\
& x_i \geq 0 \quad \forall\ i \in N_1 \\
& x_i \leq 0 \quad \forall\ i \in N_2 \\
& x_i \text{ free} \quad \forall\ i \in N_3
\end{aligned}
$$

where $c, a_i, h_i, g_i \in \mathbb{R}^n$ and N_1, N_2, N_3 are subsets of $\{1, \ldots, n\}$.

Define $A \triangleq \begin{pmatrix} — & a_1^\top & — \\ — & a_2^\top & — \\ & \vdots & \\ — & a_{m_1}^\top & — \end{pmatrix}$, $b \triangleq \begin{pmatrix} b_1 \\ \vdots \\ b_{m_1} \end{pmatrix}$

we write $Ax \geq b$. (element-wise.)

Compact Formulation

We can write LPs in a more compact way:

$$
\begin{aligned}
\text{minimize/maximize}_{x\in\mathbb{R}^n} \quad & c^\top x \\
\text{subject to} \quad & Ax \geq b \\
& Hx \leq d \\
& Gx = e \\
& x_i \geq 0 \quad \forall\ i \in N_1 \\
& x_i \leq 0 \quad \forall\ i \in N_2 \\
& x_i \text{ free} \quad \forall\ i \in N_3
\end{aligned}
$$

- A, H, and G are matrices (with dimensions $m_1 \times n$, $m_2 \times n$ and $m_3 \times n$).
- b, d, and e are vectors (with dim. $m_1 \times 1$, $m_2 \times 1$ and $m_3 \times 1$).

Standard Form of LPs

In order to study LPs more systematically, we want to have a standard form for LPs:

$$\begin{aligned} \text{minimize}_{x\in\mathbb{R}^n} \quad & c^\top x \\ \text{subject to} \quad & Ax = b \\ & x \geq 0 \end{aligned} \tag{1}$$

where $x \in \mathbb{R}^n$, A is an $m \times n$ matrix and $b \in \mathbb{R}^m$.

Remark:

- The definition of "standard form" may differ from book to book. We use (1) as the standard form in this course (consistant with Bertsimas and Tsitsiklis's book).

Transformation to Standard Forms

1. If the objective is maximization:
   - replace c with $-c$ and change it to minimization.
2. If there is inequality constraints $Ax \leq b$ or $Ax \geq b$:
   - add decision variable s, also called slack variable.
   - replace the inequality with $Ax + s = b, s \geq 0$ or $Ax - s = b, s \geq 0$.
3. If there is non-positivity constraint $x_i \leq 0$:
   - replace all x_i wtih $-y_i$, where $y_i \geq 0$.
4. If there is "free" variables x_i (no constraints on x_i):
   - replace x_i with $x_i^+ - x_i^-$, where $x_i^+ \geq 0$, $x_i^- \geq 0$

Example: Production Planning

why it is not standard form?

① max → min

② ineq. constraints → add slack variable

$$
\begin{array}{lrrl}
\text{maximize} & x_1 & +2x_2 & \\
\text{subject to} & x_1 & & \le 100 \\
& & 2x_2 & \le 200 \\
& x_1 & +x_2 & \le 150 \\
& x_1, & x_2 & \ge 0
\end{array}
$$

Standard Form:

$$
\begin{array}{lrrrrrl}
\text{minimize} & -x_1 & -2x_2 & & & & \\
\text{subject to} & x_1 & & +s_1 & & & = 100 \\
& & 2x_2 & & +s_2 & & = 200 \\
& x_1 & +x_2 & & & +s_3 & = 150 \\
& x_1, & x_2, & s_1, & s_2, & s_3 & \ge 0
\end{array}
$$

$c = \begin{pmatrix} -1 \\ -2 \end{pmatrix}$, $Ax = \begin{pmatrix} 100 \\ 200 \\ 150 \end{pmatrix}$, $A = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{pmatrix}$

Example: Support Vector Machines

why it is not standard form?

① w, b are free → represent them as difference of two non-negative terms.

② ineq. constrait → add slack variable

$$\begin{aligned} \text{minimize}_{w,b,t} \quad & \sum_i t_i \\ \text{subject to} \quad & y_i(x_i^\top w + b) + t_i \geq 1, \quad \forall\, i \\ & t_i \geq 0 \qquad\qquad \forall\, i. \end{aligned}$$

- Replace $w = w^+ - w^-$, $b = b^+ - b^-$, where w^+, w^-, b^+,$b^- \geq 0$.
- Add slack variables to eliminate inequality constraints.

$w_i = w_i^+ - w_i^-$, $w_i^+ \geq 0$, $w_i^- \geq 0$

$b_i = b_i^+ - b_i^-$, $b_i^+ \geq 0$, $b_i^- \geq 0$

$w^+ = \begin{pmatrix} w_1^+ \\ \vdots \\ w_n^+ \end{pmatrix}$ $w^- = \begin{pmatrix} w_1^- \\ \vdots \\ w_n^- \end{pmatrix}$

Standard Form for SVMs

$$
\begin{aligned}
\min_{w^+,w^-,b^+,b^-,t,s} \quad & \sum_i t_i \\
\text{subject to} \quad & y_i(x_i^\top w^+ - x_i^\top w^- + b^+ - b^-) + t_i - s_i = 1 \quad \forall i \\
& w^+, w^-, b^+, b^- \geq 0 \\
& t_i, s_i \geq 0 \qquad \forall\ i.
\end{aligned}
$$

Standard Form: Discussion

- Standard form is mainly used for analysis purposes. We do not need to write a problem in standard form unless necessary.
- Usually we just represent the problem in a way that makes it easy to understand.
- Transforming an LP into the standard form is an important skill. It is helpful for analyzing LP problems as well as when using software to solve it.

Questions?